acts by zero in E. Let us write L as

$$L = \sum_{j=0}^{r} Q_j(t) \left(\frac{d}{dt}\right)^j,$$

where $Q_r \neq 0$. Then we have

$$Lt^a = \sum_{j=0}^{r} Q_j(t) a(a-1)\dots(a-j+1)t^{a-j}.$$

This must be zero, so we have $\sum_{j=0}^{r} Q_j(t)a(a-1)\dots(a-j+1)t^{-j} = 0$ in $k[a][t,t^{-1}]$. Taking the leading term in a, we get $Q_r(t) = 0$, a contradiction.

(ii) Any word in x, y, x^{-1}, y^{-1} can be ordered at the cost of multiplying it by a power of q. This easily implies both the spanning property and the linear independence. □

Remark 2.7.2. The proof of (i) shows that the Weyl algebra A can be viewed as the algebra of polynomial differential operators in one variable t.

The proof of (i) also brings up the notion of a faithful representation.

Definition 2.7.3. A representation $\rho : A \to \operatorname{End} V$ of an algebra A is **faithful** if ρ is injective.

For example, $k[t]$ is a faithful representation of the Weyl algebra if k has characteristic zero (check it!), but not in characteristic p, where $(d/dt)^p Q = 0$ for any polynomial Q. However, the representation $E = t^a k[a][t,t^{-1}]$, as we've seen, is faithful in any characteristic.

Problem 2.7.4. Let A be the Weyl algebra.

(a) If $\operatorname{char} k = 0$, what are the finite dimensional representations of A? What are the two-sided ideals in A?

Hint: For the first question, use the fact that for two square matrices B, C, $\operatorname{Tr}(BC) = \operatorname{Tr}(CB)$. For the second question, show that any nonzero two-sided ideal in A contains a nonzero polynomial in x, and use this to characterize this ideal.

Suppose for the rest of the problem that $\operatorname{char} k = p$.